

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 5 : Session – 3 : SLO – 1 | Flynn's Taxonomy — Systems

■ I. Multiple Choice

1. b) Number of instruction streams and data streams.
2. c) SISD — the traditional single-core, single-stream machine.
3. d) SIMD — GPUs apply one instruction to many data elements.
4. c) MIMD — multicore processors run independent instruction streams per core.
5. c) MISD — multiple instruction streams on a single data stream.

■ II. True or False

1. TRUE.
2. FALSE — SIMD is one instruction stream on multiple data streams.
3. TRUE.
4. TRUE.
5. FALSE — MISD is uncommon; general-purpose systems are SISD/SIMD/MIMD.

■ III. Scenario-Based Activity

- Answer: b) Current: SIMD | Upgrade: MIMD.
- Current design: the same prediction algorithm runs on different data portions simultaneously — one instruction stream over many data streams = SIMD.
- After upgrade: different algorithms run simultaneously on different data = multiple instruction streams and multiple data streams = MIMD.